

1. How many 12 bit strings
 - a) start with 101?
 - b) start with 011 or with 0000?

2. How many hexadecimal strings of length 8
 - a) do not contain letter A?
 - b) contain at least one letter A?

 - c) contain exactly one letter A?
 - d) contain at most one letter A?

3. What value does a variable *counter* have after the following code execution?

```
counter = 100
for i = 0 to 10 {
    counter = counter + 2
}
for j = 1 to 20 {
    counter = counter + 3
    for k = 2 to 30 {
        counter = counter + 4
    }
}
```

1

```
counter = a
for i = 1 to n {
    counter = counter + 2
    for j = n to 2n {
        counter = counter + 3
        for k = 0 to 3n {
            counter = counter + 4
        }
    }
}
// assume  $n \geq 1$ 
```

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4. Consider passwords with the following restrictions:

✓ Length 12.

✓ Available characters are 26 upper case letters A, B, C... Z.

a) What is the total number of passwords?

b) How many passwords have all distinct letters?

c) How many passwords have at least one letter repeated?

d) How many passwords have four letters "A", six letters "B" and two letters "C"?

e) How many passwords have seven letters "X" and four letters "Y" (and one other letter)?

5. Consider the string COMPUTERARCHITECTURE

- a) How many arrangements are there of all the letters?

- b) How many arrangements in part a) start with HA?

- c) How many arrangements in part a) do not start with HA?

- d) How many arrangements in part a) have all vowels together?

- e) How many arrangements in part a) have all vowels together and all consonants together?

6. An alphabet of 28 symbols is used to for transmitting a message over a communication channel. Assume that each message is 10 symbols long.

a) How many different messages are there?

b) How many different messages have all symbols distinct?

c) Assuming that # is one of the symbols, how many different messages have:

i. 10 #s

ii. 9 #s

iii. 4 #s

If a message has less than 10 symbols then 29th symbol (&) is randomly inserted to make a transmission of length 10.

d) In how many different ways can the message CST be transmitted?

If all of the problems can't be done in the lab period, students can work on them on their own.